



SCSM2223- CROSS-PLATFORM APPLICATION DEVELOPMENT

Chapter 8: Individual Lab Assignment Report (Employee Directory)

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1.0 Screenshots of Working UI

1.1 Main Employee Directory Page

Employee Directory

Manage employee records for the HR office.

Total Active Employees
6

Add Employee

Employee ID
EMP001

Name
Full name

Email
name@company.com

Department
Select department

Position
Job title

Hire Date
dd/mm/yyyy

Salary RM

☐ Active employee

Add Employee

Sort by Salary

Ascending

Apply

Employee List

Employee ID	Name	Email	Department	Position	Hire Date	Salary	Status	Actions
EMP007	Nur Izzati	nur.izzati@company.com	HR	Recruitment Officer	2024-01-31	RM 3,200.00	Active	EditDelete
EMP005	Mei Ling	mei.ling@company.com	Marketing	Marketing Executive	2023-04-11	RM 3,900.00	Active	EditDelete

Figure 1.1: Screenshot of Main Employee Directory Page

1.2 Form Validation

Employee Directory

Manage employee records for the HR office.

Total Active Employees
6

Add Employee

Employee ID

E001

Employee ID must be like EMP001.

Email

jennie@gmail

Please enter a valid email.

Position

Software Engineer

Salary RM

1500

Add Employee

Name

JENNIE

Department

IT

Hire Date

23/05/2026

Hire date cannot be in the future.

☒ Active employee

Figure 1.2: Screenshot of Form Validation (Invalid Employee ID, email and hire date)

1.3 CRUD Operation

Edit Employee

Employee ID

EMP001

Email

aina.rahman@company.com

Position

Senior HR Executive

Salary RM

4000.00

Update Employee

Cancel

Name

Aina Rahman Updated

Department

HR

Hire Date

14/03/2021

☐ Active employee

Figure 1.3: Screenshot of Edit Employee Operation

1.4 Search and Sort Functionality

Sort by Employee ID

Ascending

Apply

Employee List

Employee ID	Name	Email	Department	Position	Hire Date	Salary	Status	Actions
EMP008	JENNIE	jennie@gmail.com	IT	Software Engineer	2026-04-30	RM 1,500.00	Active	EditDelete
EMP009	John	helioganier@gmail.com	IT	Software Engineer	2026-04-01	RM 4,000.00	Active	EditDelete

Figure 1.4: Screenshot of Search and Sort Functionality

2.0 Achievement of Learning Outcomes

1. LO1 - Connect Vue 3 application to an external REST API

There are API connections between the Vue 3 frontend and Express backend where the employee details are fetched dynamically and shown on the frontend. The create, update, delete, searching and sorting features are handled with the help of APIs.

2. LO2 - Configure and use the Axios HTTP client

An Axios object is created and placed in src/services/api.js instead of calling Axios directly from any components. This object contains default values such as the baseURL, request timeout, and header.

3. LO3 - Build Vue forms using v-model, validation and events

Vue 3 Composition API and v-model are used in EmployeeForm.vue component. Inline error messages are displayed in case of invalid input, while the employee form interacts with its parent components via emitted save/cancel event

4. LO4 - Build Express REST API with CRUD, search, sort, and MySQL

The Express app executes all CRUD operations by using MySQL and Prepared Statements. Server-side sorting & search functionality is supported through query parameters.

3.0 Challenges & Solutions

1. Frontend and backend running on different ports:

Vue frontend runs on port 5174, whereas the Express backend works on port 3001. It led to cross-origin communication issues, which were handled using CORS middleware in Express and Axios with the base URL being set at <http://localhost:3001>.

2. MySQL connection and local environment setup:

The backend must successfully connect to the local Laragon MySQL server before CRUD operations performed. The parameters used for connecting to the server are host '127.0.0.1', port 3306, user 'root', and database 'employee_directory'. The schema.sql file creates the database, table employees, and populates the database with seed data.

3. Secure server-side sorting functionality:

Sorting is needed for the API but letting clients sort by any random column names would be a major vulnerability. It was managed by checking the received sortBy parameter against a list of valid columns: name, hireDate, and salary.